PART IIA, PAPER 3, SUPERVISION 6
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SOLUTIONS TO PROBLEMS

S6.1 Instruments vs Panels: Dawn of justice for endogeneity of crime.
Recall S5.1, the question about the determinants of crime.

(a) Download the dataset `crime4.dta`. Describe the variables using command `des`.

(b) Observe now that we actually have a panel dataset. So we can use a first-difference or fixed effects method to eliminate omitted variable bias.

(c) Why might there be an omitted variable bias (as opposed to simultaneity bias) in a regression of crime on police?
Solution: Places with high unemployment may have higher crime rates, and police jobs are attractive options since there aren't many jobs around for the youth.

(d) Regress `lcrime` on `lpolice ldensity urban west central`. What is the interpretation of the coefficient on `lpolice`?
Solution: Elasticity of crime with respect to police holding other things fixed (partial effect).

(e) Run a first-differenced model by regressing `clcrime` on `clpolice cldensity`.

(f) How does your first-differenced estimate of the effect of compare with that from the IV regression?
Solution: Still positive and significant.

(g) Is `polpc` strictly exogenous? If not, how does that affect the estimates?
Solution: If more crime last period leads to more police this period, then strict exogeneity is violated. We will mistakenly conclude that police increase has positive effect on crime. However, the significance positive effect may in fact have some empirical basis, and interesting read is Philip Alston's report for the UN on extreme poverty and human rights in the US. Among the most interesting observations is that the "*so-called 'fines and fees' are piled up so that low level infractions become immensely burdensome, a process that affects only the poorest members of society who pay the vast majority of such penalties. State, county and municipal police and law enforcement agencies are not always forces for change in such settings.*"
Solution: Here is `STATA` code for this the whole problem

```
// OLS regression
reg crmrte polpc west central urban

// IV regression
ivregress 2sls crmrte west central urban (polpc = taxpc), first

// Wu-Hausman and Durbin test for endogeneity of polpc
estat endogenous
```

```
// Test endogeneity of polpc manually
reg polpc taxpc west central urban
predict res, res
reg crmrte polpc west central urban res

// Regression for the logs
reg lpolpc ldensity urban west central

// First-differenced regression
reg clcrmrte clpolpc cldensity
```

S6.2 Get your maximum likelihood out if approached by a coin-tossing hustler. Suppose we want to test if a coin is fair. We toss the coin 10 times and get the following outcomes: $HHTTHTHHTH$.

(a) Write down the likelihood function in terms of the probability of H in one toss.

Solution: Denote the outcome of a coin toss indexed by i with X_i, so that $X_i \sim \text{Ber}(p)$ for $i = 1, \ldots, n$, and some probability of observing a head given by p. Then $\mathbb{P}(X_i = 1|p) = p$ and $\mathbb{P}(X_i = 0|p) = 1 - p$. The probability mass function of X_i, for a given value of p, is

$$f(X_i; p) = p^{X_i}(1-p)^{1-X_i}. \tag{1}$$

Then, the likelihood function $L(p|\mathbf{X})$ is given by the the joint probability of observing $\mathbf{X} = (X_1, \ldots, X_n)$, denoted by $f(\mathbf{X}; p)$, provided that these random variables are independent and have the same mass function $f(X_i; p)$:

$$L(p; \mathbf{X}) = f(\mathbf{X}; p) = \prod_{i=1}^{n} f(X_i; p) = \prod_{i=1}^{n} p^{X_i}(1-p)^{1-X_i} \tag{2}$$

Note the we can also express the likelihood function in terms of the number of observed heads in a sample of n coin tosses, denote this number by it by n_1, and the number of tails, denote it by $n - n_1$:

$$L(p; \mathbf{X}) = f(\mathbf{X}; p) = p^{n_1}(1-p)^{n-n_1}. \tag{3}$$

(b) What is the probability of obtaining the above sample if the probability of H in a single draw was 0.5?

Solution: The observed sample of size $n = 10$ is given by $\mathbf{x} = (1, 1, 0, 0, 1, 0, 1, 1, 0, 1)$, where the number of heads is $n_1 = 6$, and the number of tails is $n - n_1 = 4$. If $p = 0.5$, then the joint probability of observing $\mathbf{x}$ is

$$f(\mathbf{x}; p = 0.5) = \prod_{i=1}^{n} \left(\frac{1}{2}\right)^{x_i} \left(\frac{1}{2}\right)^{1-x_i} = \left(\frac{1}{2}\right)^{6} \left(\frac{1}{2}\right)^{4} = 9.76 \times 10^{-4}. \tag{4}$$

(c) Write down the log-likelihood function and find the maximum likelihood estimator.

Solution: Take the log of $L(p|\mathbf{X})$

$$l(p; \mathbf{X}) = \ln L(p; \mathbf{X}) = \sum_{i=1}^{n} [X_i \ln p + (1 - X_i) \ln(1-p)] \tag{5}$$

To find the maximum likelihood estimate of p, denoted by $\hat{p}$, given the data $\mathbf{X}$, compute the first-order derivative of $l(p|\mathbf{X})$ with respect to p and set it equal to zero:

$$\frac{\partial}{\partial p} l(p;\mathbf{X}) = \sum_{i=1}^{n} \left[\frac{X_i}{p} - \frac{1-X_i}{1-p} \right] = 0 \implies \hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i. \tag{6}$$

so the estimator is the sample mean. Check that this is indeed the maximum of the likelihood function

$$\frac{\partial^2}{\partial p^2} l(p;\mathbf{X}) = \sum_{i=1}^{n} \left[-\frac{X_i}{p^2} - \frac{(1-X_i)}{(1-p)^2} \right] < 0 \tag{7}$$

for all value of p, and hence $\hat{p}$ is indeed the maximum of the log-likelihood function $l(p;\mathbf{X})$. For the observed data with six heads and four tails, the estimate is then the proportion of heads, namely $\hat{p} = 0.6$.

(d) Can you estimate the variance of this estimator?

Solution: Recall that the variance of a Bernoulli random variable X_i is given by $\mathrm{Var}(X_i) = p(1-p)$. Therefore, the variance of the maximum likelihood estimator is given by

$$\mathrm{Var}(\hat{p}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{p(1-p)}{n},$$

since X_i are independent and so $\mathrm{Cov}(X_i, X_j) = 0$ for all $i \neq j$. The estimate of the variance for our observed sample is therefore based on $\hat{p}$ rather than the unobserved parameter p, and equals

$$\mathrm{Var}(\hat{p}) = \frac{\hat{p}(1-\hat{p})}{n} = \frac{0.6 \times 0.4}{10} = 0.024.$$

(e) What is the probability of obtaining the above sample if the true probability of H in a single draw was equal to the maximum likelihood estimate? Compare your results in (b) and (f) and comment on the fairness of the coin.

Solution: Assuming that $p = \hat{p} = 0.6$, the probability of observing the above sample with 6 heads and 4 tails is

$$f(\mathbf{x}; p = 0.6) = 0.6^6 \times 0.4^4 = 1.19 \times 10^{-3} > 9.76 \times 10^{-3} f(\mathbf{x}; p = 0.5)$$

Conclude that the sample is more likely for an unfair coin.

(f) Construct and perform a hypothesis test for testing the fairness of the coin.

Solution: To assess the fairness of the coin we want to test $H_0 \colon p = p_0 = 0.5$ against $H_1 \colon p \neq 0.5$, at say 5% significance. Since the sample size $n = 10$ is too small for the central limit theorem to apply, the following test is only a very rough approximation. In large samples, we know that under the null

$$\hat{p} \overset{a}{\sim} N\left(p_0, \frac{p_0(1-p_0)}{n}\right) \iff T = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \overset{a}{\sim} N(0,1).$$

Reject the null at 5% if the observed test statistic t is such that $|t| > 1.96$.

$$t = \frac{0.6 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{10}}} = 0.632 < 1.96$$

Do not reject the null at 5%, conclude that there is not sufficient evidence to suggest that the coin is unfair.

S6.3 Do kids even watch TV these days?
We will consider the determinants of childhood obesity.

(a) Download the dataset `obesekids.dta`.

(b) See what the variables mean by typing `des`.

(c) Summarize the data using the command `summ`.

(d) Run a probit regression with the following command:
`probit obesec ageyrs female white black hisp tvyest povrat`

(e) What do you infer from the row corresponding to `hisp`.
Solution: relative to Asians (non-white, non-black and non-Hispanic), Hispanic kids have a higher chance of being obese, given the same age, gender, poverty and TV watched.

(f) Compute the predicted probability of being obese for a 16 year old female white child who watched 3 hours of TV, whose family is income-to-needs ratio is 4.56.
Solution: The probit predicted probability of $\widehat{\mathbb{P}}(\texttt{obesec}_i = 1 \mid \mathbf{x}_i)$, where $\mathbf{x}_i$ contains the explanatory variables with values given in the question are given by the standard normal cdf:

$$\Phi(-1.56 + 0.0076 \times 16 + 0.029 \times 1 + 0.023 \times 1 + 0.056 \times 3 + 0.0019 \times 4.56)$$

The z-score is equal to -1.206 and hence the predicted probability of being obese for a 16 year old female white child who watched 3 hours of TV, whose family is income-to-needs ratio is 4.56 is $\Phi(-1.206) = 0.1139$.

(g) How much would this probability fall for an identical person as above but who watched no TV yesterday? Explain whether the fall is statistically significant?
Solution: Find the same predicted probability as above, but now set TV hours to zero:

$$\Phi(-1.56 + 0.0076 \times 16 + 0.029 \times 1 + 0.023 \times 1 + 0.056 \times 0 + 0.0019 \times 4.56)$$

The z-score is -1.376 and the corresponding predicted probability is 0.0844, so the drop is 0.0295, which is statistically significant, given that `tvyest` has a significance coefficient.

(h) How would you test that race has no effect on the probability of being obese? Now test it using the corresponding `STATA` command. What do you conclude?
Solution: Conduct a test that dummies white, black and Hispanic are jointly zero

```
test white black hisp
```

This is a likelihood ratio test, the test statistic has a χ^2_3 asymptotic distribution. The sample test statistic is 19.69 with a p-value of 0.0002, so reject the null with very low significance levels. The race dummies are jointly significant.

(i) What does your overall analysis tell us about whether we should discourage young children from watching TV in order to prevent them from being obese?

Solution: Some discussion of omitted variables and reverse causality - reasons why obese status may be the cause of more TV watching.

ADDITIONAL PROBLEMS

S6.4 There are plenty more fish in the sea cleaned and estimated by probit.
A researcher studies the effect of water pollution on the mortality rate in a fish population. He runs experiments and determines that the 0.1% and 0.05% concentrations of a pollutant in water lead to 50% and 23% mortality rate, respectively.

(a) Derive the probit model specification for the mortality rate (probability of death described by the normal distribution) from the auxiliary regression for some appropriate latent variable.

Solution: Denote concentration of pollutant by X_i and the binary death/no death outcome for fish i by D_i, where D_i is derived from some latent variable D_i^* such that

$$D_i = \begin{cases} 1 & \text{if } D_i^* > 0 \\ 0 & \text{if } D_i^* \leq 0 \end{cases}$$

Suppose that the model for the latent variable is linear in and depends on X_i, so that

$$D_i^* = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \ldots, n, \tag{8}$$

The crucial assumption of the probit model is that $u_i \sim \text{iid}N(0,1)$. We can then derive the probabilistic model for D_i conditional on X_i as follows

$$\begin{aligned}
\mathbb{P}(D_i = 1|X_i) &= \mathbb{P}(D_i^* > 0|X_i) \\
\text{[sub-in latent model]} &= \mathbb{P}(\beta_0 + \beta_1 X_i + u_i > 0|X_i) \\
\text{[re-arrange for } u_i\text{]} &= \mathbb{P}(u_i > -\beta_0 - \beta_1 X_i|X_i) \\
\text{[property of probaiblity]} &= 1 - \mathbb{P}(u_i \leq -\beta_0 - \beta_1 X_i|X_i) \\
[u_i \text{ standard normal, so symmetrical}] &= \mathbb{P}(u_i \leq \beta_0 + \beta_1 X_i|X_i) \\
&= \Phi(\beta_0 + \beta_1 X_i)
\end{aligned}$$

where $\Phi(\cdot)$ is the cdf of the standard normal random variable u_i. More formally, the probit model for D_i can be expressed as a function of the realization $X_i = x$:

$$\mathbb{P}(D_i = 1|X_i = x) = \Phi(\beta_0 + \beta_1 x). \tag{9}$$

(b) Assuming that the probability of death is described by a probit model, determine the mortality rate of fish in absolutely clean water.

Answer: We have two data points and the corresponding predicted probabilities, which we can use to determine the estimates of β_0 and β_1, namely

$$\widehat{\mathbb{P}}\left(D_i = 1|X_i = 0.001\right) = \Phi\left(\hat{\beta}_0 + \hat{\beta}_1 \times 0.001\right) = 0.5$$
$$\widehat{\mathbb{P}}\left(D_i = 1|X_i = 0.0005\right) = \Phi\left(\hat{\beta}_0 + \hat{\beta}_1 \times 0.0005\right) = 0.23$$

Use the fact that $\Phi^{-1}(0.5) = 0$ and $\Phi^{-1}(0.23) = -0.739$ to show that

$$\hat{\beta}_0 + \hat{\beta}_1 \times 0.001 = 0$$
$$\hat{\beta}_0 + \hat{\beta}_1 \times 0.0005 = -0.739$$

Solving this system of simultaneous equations yields $\hat{\beta}_0 = -1.48$ and $\hat{\beta}_1 = 14.8$. Finally, the probability of death, or in other words the mortality rate in clean water, that is probability that $D_i = 1$ conditional on $X_i = 0$ is given by

$$\widehat{\mathbb{P}}\left(D_i = 1|X_i = 0\right) = \Phi\left(\hat{\beta}_0 + \hat{\beta}_1 \times 0\right) = \Phi\left(-1.48\right) = 0.0694.$$

Therefore the mortality rate in clean water is 6.94%.

(c) Let $\Phi(x)$ be the cdf of $N(0,1)$. The values of the standard normal density at x_1 and x_2 such that $\Phi(x_1) = 0.5$ and $\Phi(x_2) = 0.23$ are 0.399 and 0.303, respectively. Using this information, compare the marginal effects of reducing the concentration of the pollutant from the initial values of 0.1% and 0.05%.

Answer: Recall that unlike for a linear model, marginal effects implies by the probit model differ for various observations of the explanatory variable. For instance, the marginal effect of X_i on the probability that $D_i = 1$ is given computed as

$$\mathrm{ME}\left(X_i\right) = \frac{\partial}{\partial X_i}\widehat{\mathbb{P}}\left(D_i = 1|X_i\right) = \frac{\partial}{\partial X_i}\Phi\left(\hat{\beta}_0 + \hat{\beta}_1 X_i\right) = \hat{\beta}_1 \phi\left(\hat{\beta}_0 + \hat{\beta}_1 X_i\right), \qquad (10)$$

where $\phi(x) = \frac{d}{dx}\Phi(x)$ is the pdf of the standard normal random variable. We are told that for x_1 such that $\Phi(x_1) = 0.5$, the pdf is given by $\phi(x_1) = 0.399$, and for x_2 such that $\Phi(x_2) = 0.23$, the pdf is given by $\phi(x_2) = 0.303$. We can use these to compute the ratio of marginal effects:

$$\frac{\mathrm{ME}\left(X_i = 0.001\right)}{\mathrm{ME}\left(X_i = 0.0005\right)} = \frac{\hat{\beta}_1 \phi\left(\hat{\beta}_0 + \hat{\beta}_1 0.001\right)}{\hat{\beta}_1 \phi\left(\hat{\beta}_0 + \hat{\beta}_1 0.0005\right)} = \frac{0.399}{0.303} = 1.317.$$

This means that the marginal effect of reducing pollution will be 1.317 higher for pollution concentration levels of 0.1% than for 0.05% levels.

(d) Discuss potential pitfalls of the linear probability model for this type of problems.

Answer: The linear probability model does not use the latent variable and directly specifies the following regression:

$$D_i = \beta_0 + \beta_1 X_i + u_i.$$

Predictions of a linear probability model can lie outside the $[0, 1]$ interval of probability values. For, example solving for the linear coefficients with the given data points:

$$\hat{\beta}_0 + \hat{\beta}_1 \times 0.001 = 0.5$$
$$\hat{\beta}_0 + \hat{\beta}_1 \times 0.0005 = 0.23$$

implies that $\hat{\beta}_0 = -0.04$, which means that linear probability would predict that the mortality rate in clean water is -4%. This prediction is meaningless.

S6.5 An opportunity to read about Norman Borlaug and the Green Revolution. A production function for farms is

$$\log Y_{it} = \alpha + \beta \log X_{it} + \gamma \log S_{it} + u_{it}, \tag{11}$$

where Y_{it} is the crop produced at farm i during period t, X_{it} is the amount of labour employed, S_{it} describes the quality of soil, and u_{it} consists of random factors such as weather. Suppose that you observe Y_{it} and X_{it} for $i = 1, \ldots, n$ and $t = 1, \ldots, T$, but that you do not observe S_{it}.

(a) Under what conditions is the OLS estimator of β, from the regression of Y_{it} on a constant and X_{it}, consistent for β? Are these conditions plausible from the microeconomic perspective? Explain.

Solution: Since S_{it} are unobserved, we can only estimate the following regression

$$\log Y_{it} = \alpha + \beta \log X_{it} + \varepsilon_{it}$$

For consistency of $\hat{\beta}_{\text{OLS}}$, we need exogeneity of X_{it} with respect to ε_{it}. This is likely to be violated due to the omitted variable S_{it} that affects produced crop and is correlated with the amount of labour employed, which makes the error term in the estimated regression, which we can express as $\varepsilon_{it} = \gamma \log S_{it} + u_{it}$, depend on X_{it}. The variables X_{it} and S_{it} are likely to be correlated since labour employed will depend on the quality of soil, as farmers maximize their profits.

(b) Describe the fixed effects estimator of β. Is it consistent? Explain.

Solution: Using the within transformation, estimate the new regression

$$\tilde{Y}_{it} = \beta \tilde{X}_{it} + \tilde{u}_{it}. \tag{12}$$

The "tilde" variables are defined as the deviations from the respective time averages:

$$\tilde{Y}_{it} = \log Y_{it} - \frac{1}{T} \sum_{i=1}^{T} \log Y_{it}$$
$$\tilde{X}_{it} = \log X_{it} - \frac{1}{T} \sum_{i=1}^{T} \log X_{it}$$
$$\tilde{S}_{it} = \log S_{it} - \frac{1}{T} \sum_{i=1}^{T} \log S_{it}$$
$$\tilde{u}_{it} = \log u_{it} - \frac{1}{T} \sum_{i=1}^{T} \log u_{it}$$

Note that the estimator of β from the above fixed-effects regression will be consistent for β as long the quality of soil is time invariant, $S_{it} = \bar{S}_i$, which is likely to be the case, especially in the short to medium run, since the fixed effects are excluded from the model and potential endogeneity is eliminated.

(c) It was observed that producing an excessively large quantity of the crop during period t leads to lower productivity of plants during the next period. How would you modify equation (11) to model this phenomenon?

Solution: Define a dummy variable for excessive crop production at time t

$$D_{i,t} = \begin{cases} 1 & \text{execess production in period } t \\ 0 & \text{otherwise} \end{cases}$$

Add the lag of $D_{i,t}$ as an explanatory variable in the original regression:

$$\log Y_{i,t} = \alpha + \gamma D_{i,t} + \beta \log X_{i,t} + \gamma \log S_{i,t} + u_{i,t}.$$

Then γ will show the effect of excessive crop yield in the previous period.

S6.6 Children are like wages, they also suffer from endogeneity of education. The `STATA` dataset `fertility.dta` contains demographic information for women in Botswana during 1988, including the number of children, years of education, age, religious and economic status variables.

```
>> eststo clear
>> eststo:  quietly reg children educ age agesq
>> eststo:  quietly reg educ age agesq frsthalf
>> eststo:  quietly reg ivreg children age agesq (educ = frsthalf
>> eststo:  quietly reg children educ age agesq tv electric bicycle
>> eststo:  quietly reg children age agesq tv electric bicycle (educ
= frsthalf)
```

(a) Estimate the following linear model using OLS

$$\texttt{children}_i = \beta_0 + \beta_1 \texttt{educ}_i + \beta_2 \texttt{age}_i + \beta_3 \texttt{age}_i^2 + u_i, \quad i = 1, \ldots, n. \tag{13}$$

Holding age fixed, compute the estimated effect of another year of education on fertility. If 100 women receive another year of education, how many fewer children are they expected to have?

Solution: Run the following code and report output in column (2) of Table 1

```
>> reg children educ age agesq
```

Conclude that an extra year of education, holding age fixed, results in 0.091 fewer children. This effect is highly significant. For a group of 100 women, if each gets another year of education, they are collectively predicted to have about 9 fewer children.

(b) The variable $\texttt{frsthalf}_i$ is a dummy variable equal to one if the woman was born during the first six months of the year. Assuming that $\texttt{frsthalf}_i$ is uncorrelated with the error term u_i, suggest and estimate a regression to show that $\texttt{frsthalf}_i$ is a reasonable IV candidate for $\texttt{educ}_i$.

Solution: Run the reduced form regression for $\texttt{educ}_i$

$$\texttt{educ}_i = \pi_0 + \pi_1 \texttt{age}_i + \pi_2 \texttt{age}_i^2 + \pi_3 \texttt{frsthal}f_i + v_i, \tag{14}$$

using the following code and report the result in column (1) of Table 1

```
>> ivreg reg educ age agesq frsthalf
```

We require $\pi_3 \neq 0$ for identification. The regression output suggests $\widehat{\pi}_3 \approx -0.852$ with the corresponding t-stat of -7.55 and so we reject H_0 of insignificance. The identification assumption seems to hold. Thus, women born in the first half of the year are predicted to have almost one year less of education, holding age fixed. In Botswana there is (or was during the relevant time period) a regulation that makes those born in the first half of the year enter school later.

(c) Estimate the model in (13) using $\texttt{frsthalf}_i$ as an IV for $\texttt{educ}_i$. Compare the estimated effect of education from the IV regression with the one obtained using OLS in (a). Explain why these estimates are different.

Solution: Use the following code and report the result in column (3) of Table 1

```
>> ivreg children age agesq (educ = frsthalf)
```

The estimated effect of education on fertility is now much higher than the corresponding OLS estimate from column (2). However, the standard error is much higher too. Indeed, we cannot reject the hypothesis that the IV estimate is equal to the OLS estimate thereby implying that the difference between OLS and IV is due to sampling error (you can check this using the Hausman test).

(d) Add the binary variables $\texttt{electric}_i$, $\texttt{tv}_i$ and $\texttt{bicycle}_i$ to the model in equation (13). Assume these variables are exogenous. Estimate the new model by OLS and 2SLS (using $\texttt{frsthalf}_i$ as an IV for $\texttt{educ}_i$). Compare the estimated coefficients on $\texttt{educ}_i$. Interpret the coefficient on $\texttt{tv}_i$ and explain why television ownership has a negative effect on fertility.

Solution: Use the following code and report the OLS results in column (4) and the IV results in column (5) of Table 1

```
>> reg children educ age agesq tv electric bicycle
>> ivreg children age agesq tv electric bicycle (educ = frsthalf)
```

Adding `electric`, `tv`, and `bicycle` to the model reduces the magnitude of the estimated effect of `educ` on the number of children in both cases, but not by too much. Note that $\hat{\beta}_{\texttt{tv}}^{OLS} = -0.253$ and is highly significant, which may be because it acts as a proxy for income or geographical location (rural vs urban). Interestingly, the effect of TV ownership is insignificant in the IV equation, $\hat{\beta}_{\texttt{tv}}^{IV} = -0.0026$, and the coefficient on `electric` is also greatly reduced in magnitude. The substantial reductions in these coefficients suggest that a linear model may not be the correct functional form, which should not be surprising since `children` is a count variable, not a continuous variable. Another explanation is that TV reduces fertility through the provision of information, in which case it might not have any explanatory power once we control for education (in an appropriately treating its own endogeneity using an instrument).

Table 1: `STATA` output for OLS and IV regressions

	(1) OLS `educ`	(2) OLS `child`	(3) IV `child`	(4) OLS `child`	(5) IV `child`
`age`	-0.108^{*}	0.332^{***}	0.324^{***}	0.340^{***}	0.328^{***}
	(2.57)	(20.09)	(18.12)	(20.69)	(17.22)
`agesq`	-0.0005	-0.0026^{***}	-0.0027^{***}	-0.0027^{***}	-0.0027^{***}
	(-0.73)	(-9.65)	(-9.55)	(-10.01)	(-9.84)
`frsthalf`	-0.852^{***}				
	(-7.55)				
`educ`		-0.0906^{***}	-0.171^{**}	-0.0767^{***}	-0.164^{*}
		(-15.30)	(-3.22)	(-12.08)	(-2.50)
`tv`				-0.253^{**}	-0.0026
				(-2.77)	$(-0.01))$
`electric`				-0.303^{***}	-0.107
				(-3.97)	(-0.64)
`bicycle`				0.318^{***}	0.332^{***}
				(6.44)	(6.44)
`const`	9.693^{***}	-4.138^{***}	-3.388^{***}	-4.390^{***}	-3.591^{***}
	(16.21)	(-17.20)	(-6.18)	(-18.27)	(-5.57)
N	4361	4361	4361	4356	4356
R^2	0.108	0.569	0.550	0.576	0.558
adj. R^2	0.107	0.568	0.550	0.575	0.557
F-stat	175.2	1915.2	1765.1	984.9	921.7
SSR	60001.1	9284.1	9682.2	9116.1	9511.7
RMSE	3.711	1.460	1.491	1.448	1.479

t statistics in parentheses

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

S6.7 Trump, Mohamed Ali and Elvis Presley did not part of this question.
A researcher interested in the effect of military service on human capital collects data from a random sample of $n = 4000$ workers aged 40 and runs the OLS regression

$$Y_i = \alpha + \beta X_i + \sum_{j=1}^{k} \gamma_j Z_{ji} + u_i, \quad i = 1, \ldots, n \tag{15}$$

where Y_i is the worker's annual earnings, Z_{ji}, $j = 1, \ldots, k$ represent k measures of observed characteristics for individual i, and X_i is a binary variable that is equal to 1 if the person served in the military and 0 otherwise. α, β and γ_j, $j = 1, \ldots, k$ are unknown parameters. We assume $\mathrm{Cov}\,(Z_{ji}, u_i) = 0$ for $j = 1, \ldots, k$.

(a) Explain why the OLS estimator of β is likely to be unreliable.

Solution: The OLS estimator of β is likely to be biased and inconsistent because X_i is likely to be endogenous, meaning that the value of the unobserved component u_i depends of whether the person served in the military or not, so that $\mathbb{E}[u_i|X_i] \neq 0$. Exogeneity of X_i will be violated since the military service status is likely to be correlated with other factors affecting income of individuals (which are reflected in u_i). These factors could include innate ability, parental income and other socio-economic parameters, or the motivation to earn money. All of these are incorporated in the unobserved error term u_i. As a results, there will be endogeneity (implicitly caused by omitted variables) in the model, causing the OLS estimator $\hat{\beta}_{\text{OLS}}$ to be biased. From economic theory, we would expect $\text{Cov}(X_i, u_i) < 0$, meaning that military status is negatively correlated with the unobserved parameters that (positively) affect income. Therefore, the OLS estimate is likely to underestimate the true effect of military service on income, $\hat{\beta}_{\text{OLS}} < \beta$.

(b) During the Vietnam War there was a draft, where priority for the draft was determined by a national lottery. Birth dates were randomly selected and ordered 1 through 365. Those with birth date ordered first were drafted before those with birth date ordered second, and so forth. Explain how the lottery might be used as an instrument to estimate the effect of service on earnings. Provide full detail of the estimation method and show that the suggested estimator is consistent.

Solution: Denote day of birth by $\texttt{db}_i$, then for a given draft year, a variable $D_i = \mathbf{1}(\texttt{db}_i < \tau)$ is constructed to determine who is eligible for the draft:

$$D_i = \begin{cases} 1 & \text{if } \texttt{db}_i < \tau \implies \text{draft eligible} \\ 0 & \text{if } \texttt{db}_i \geq \tau \implies \text{draft ineligible} \end{cases}$$

The integer τ was chosen based upon the demand for conscripts in a given year. Setting $D_i = 1$ indicates that the i'th individual is draft-eligible, and $D_i = 0$ if he is ineligible. It is proposed to use D_i as an instrument for X_i in regression 15. We want the instrument to be randomly assigned across individuals. This is definitely the case with D_i, as the draft lottery was random. Second, we want the instrument to be valid (exclusion restriction), or uncorrelated with the residuals in the original regression, $\mathbb{E}[u_i|D_i] = 0$. Since the draft was random, it is highly unlikely that the draft eligibility status is correlated with the unobserved parameters that affect income, such as innate ability or the environmental development factors entering u_i. Thirdly, we want the instrument to be relevant (informative), that is correlated with the endogenous explanatory variable $\text{Cov}(D_i, X_i) \neq 0$. This condition also holds, since draft-eligible individuals were clearly more likely to serve in the military and so $\text{Cov}(D_i, M_i) > 0$.

We can use a two-stage least squares to estimate β. In the first-stage, the following reduced-form regression of X_i on D_i is estimated with OLS:

$$X_i = \pi_0 + \pi_1 D_i + \sum_{j=1}^{k} \delta_j Z_{ji} + v_i, \quad i = 1, \ldots, n, \tag{16}$$

where v_i are the error terms. We assume that D_i is exogenous with respect to v_i, that is $\mathbb{E}[v_i|D_i] = 0$, and that D_i is informative for X_i, that is $\text{Cov}(D_i, X_i) \neq 0$. The

OLS estimator of π_1 from (16) is given by

$$\hat{\pi}_1 = \frac{\widehat{\text{Cov}}\left(D_i, X_i\right)}{\widehat{\text{Var}}\left(D_i\right)}. \tag{17}$$

The corresponding predicted values of X_i from the first-stage regression is then computed using $\widehat{X}_i = \hat{\pi}_0 + \hat{\pi}_1 D_i + \sum_{j=1}^{k} \hat{\delta}_j Z_{ji}$, for all $i = 1, \ldots, n$. In the second-stage, estimate the following reduced-form regression:

$$Y_i = \alpha + \beta \hat{X}_i + \sum_{j=1}^{k} \gamma_j Z_{ji} + u_i, \quad i = 1, \ldots, n. \tag{18}$$

In a simple case with no additional exogenous variables, the IV estimator of β from regression (18), denoted by $\hat{\beta}_{\text{IV}}$, is then given by

$$\hat{\beta}_{\text{IV}} = \frac{\widehat{\text{Cov}}\left(\widehat{X}_i, Y_i\right)}{\widehat{\text{Var}}\left(\widehat{X}_i\right)}. \tag{19}$$

Consider the numerator and the denominator separately:

$$\begin{aligned}
\widehat{\text{Cov}}\left(\widehat{X}_i, Y_i\right) &= \widehat{\text{Cov}}\left(\hat{\pi}_0 + \hat{\pi}_1 D_i, Y_i\right) = \hat{\pi}_1 \widehat{\text{Cov}}\left(D_i, Y_i\right), \\
\widehat{\text{Var}}\left(\widehat{X}_i\right) &= \widehat{\text{Var}}\left(\hat{\pi}_0 + \hat{\pi}_1 D_i\right) = \hat{\pi}_1^2 \widehat{\text{Var}}\left(D_i\right).
\end{aligned}$$

Putting these two together in expression (19) yields

$$\hat{\beta}_{\text{IV}} = \frac{\widehat{\text{Cov}}\left(\widehat{X}_i, Y_i\right)}{\widehat{\text{Var}}\left(\widehat{X}_i\right)} = \frac{\hat{\pi}_1 \widehat{\text{Cov}}\left(D_i, Y_i\right)}{\hat{\pi}_1^2 \widehat{\text{Var}}\left(D_i\right)} = \left(\frac{1}{\hat{\pi}_1}\right)\left(\frac{\widehat{\text{Cov}}\left(D_i, Y_i\right)}{\widehat{\text{Var}}\left(D_i\right)}\right).$$

Substitute for $\hat{\pi}_1$ from expression in (17) to conclude

$$\hat{\beta}_{\text{IV}} = \left(\frac{\widehat{\text{Var}}\left(D_i\right)}{\widehat{\text{Cov}}\left(D_i, M_i\right)}\right)\left(\frac{\widehat{\text{Cov}}\left(D_i, Y_i\right)}{\widehat{\text{Var}}\left(D_i\right)}\right) = \frac{\widehat{\text{Cov}}\left(D_i, Y_i\right)}{\widehat{\text{Cov}}\left(D_i, M_i\right)} = \beta + \frac{\widehat{\text{Cov}}\left(D_i, u_i\right)}{\widehat{\text{Cov}}\left(D_i, M_i\right)} \tag{20}$$

Since by the law of large numbers, as $n \to \infty$, $\widehat{\text{Cov}}\left(D_i, u_i\right) \to \text{Cov}\left(D_i, u_i\right) = 0$, by assumption of exogeneity of the instrument, we can conclude that $\hat{\beta}_{\text{IV}} \to \beta$ and the IV estimator is consistent for β.

(c) Would you expect the R^2 from the IV regression based on the instrument from (b) to be larger or smaller than the R^2 from the OLS regression? Explain.

Solution: We would expect R^2 from the IV regression to be lower, since in the second stage of the 2SLS procedure only exogenous variability in X_i is included, as expressed by $\hat{X}_i$, while in the OLS regression it is the full variability of X_i.